



Future Agricultural Engineer |
Precision Agriculture & Data Specialist

CONTACT

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EDUCATION

B.S. IN AGRICULTURAL ENGINEERING (AGRONOMIA)
UNIVERSIDADE FEDERAL DE VICOSA(UFV)
(FEDERAL UNIVERSITY OF VICOSA)
EXPECTED GRADUATION: AUG/2026

SKILLS & CORE

- PROGRAMMING & CORE TECH: PYTHON (FASTAPI, ASYNCIO, PANDAS, NUMPY, XGBOOST, SCIKIT-LEARN), RUST, MACRO/VBA, APACHE SPARK
- DATA ENGINEERING & CLOUD: APACHE NIFI (ETL/ELT FLOWS), DATABRICKS (DISTRIBUTED PROCESSING), AWS (EC2, S3, CLOUD ARCHITECTURE), MONGODB (NOSQL AGGREGATION PIPELINES).
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- AGTECH & GEOTECHNOLOGIES: PRECISION AGRICULTURE SYSTEMS, DIGITAL AGROPECUÁRIA, MQTT PROTOCOL, QGIS, GOOGLE EARTH ENGINE.
- DEVOPS & DEPLOY: DOCKER, RENDER, VERCEL, STREAMLIT.

ADDITIONAL INFORMATION

LANGUAGES:

- ENGLISH (INTERMEDIATE - TECHNICAL WRITING AND DOCUMENTATION)
- SPANISH (INTERMEDIATE).

CORE PROFESSIONAL QUALITIES:

- HIGH AUTONOMY FOR REMOTE WORKFLOWS
- STRUCTURED PROBLEM-SOLVING,
- CLEAN CODE PRINCIPLES

ARLEU JÚNIOR

Agronomy student at the Federal University of Viçosa (expected graduation aug/2026) specializing in the intersection of Precision Agriculture, Asynchronous Systems, and Data Engineering. Combines deep field operations insight with a strong backend stack to design real-time data ingestion pipelines, cloud-hosted microservices, and statistical data quality engines. A highly autonomous and adaptable problem solver with practical experience managing distributed IoT data architectures (Python, FastAPI, SQL), dedicated to engineering scalable cloud solutions that optimize complex agribusiness operations.

PROFESSIONAL EXPERIENCE

Pif Paf Alimentos

Field Production & Data Intern (2025 – 2026)

- Monitored commercial poultry operations alongside the technical coordination team, collecting field metrics regarding sanitary voids, bio-security protocols, and housing preparation.
- Conducted biological data sampling, field necropsies, and pre-slaughter visual assessments to track flock health and operational compliance.
- End-to-End Software & Data Engineering: Architected and deployed a functional MVP (Minimum Viable Product) for a Predictive Web Application (Poultry Weight Prediction) utilizing Python, Streamlit, and Pandas. Engineered robust data pipelines with built-in sanity checks to process time-series field data, integrating an XGBoost regression model (via One-Hot Encoding) to prototype automated slaughter weight forecasting.
- Participated in weekly technical coordination meetings, translating field observations into operational insights for production efficiency.

ADWA Cannabis – Protected Cultivation and Forest Nursery

Field Data & Operations Intern (2022 - 2025)

- Monitored environmental and crop metrics within protected cultivation facilities (greenhouses) and forest nurseries, focusing on data-driven quality control of plant materials.
- Applied technical protocols for vegetative propagation (cutting, cloning), fertilization, and phytosanitary management, tracking operational variables to ensure consistency.
- Supported harvest and post-harvest operations, implementing data logging workflows to guarantee strict process standardization and product traceability.
- Participated in applied agronomic research, leading the collection, systemization, and analysis of field datasets to support high-level technical decision-making.

GeCotton – Cotton Study Group (UFV)

Events Assessor (2025 – Present)

- Organized technical and scientific events focused on innovation and technology transfer, bridging the gap between academia, regional producers, and enterprises.
- Developed and implemented a script/tool (Nomino) during event operations to automate the generation and signing of certificates, optimizing administrative workflows.
- Collaborated on technical materials, scientific dissemination, and training sessions covering machinery management and precision agriculture.
- Partnered with researchers and industry professionals, supporting experimental field operations and technical data collection.



ARLEU JÚNIOR

SELECTED TECH & DATA CERTIFICATIONS

Databricks Academy (2026)

- Get Started with Data Engineering,
- Building Data Pipelines (ETL/ELT),
- Autonomous AI Agents Architecture.

AWS Skill Builder (2026)

- AWS Technical Essentials,
- Cloud Quest: Cloud Practitioner,
- AWS Technical Essentials for AgriTech.

MongoDB University (2026)

- CRUD Operations,
- MongoDB Aggregation Fundamentals (ETL pipelines),
- Relational to Document Data Modeling.

Apache NiFi

- Fundamentals,
- Data Flows Crash Course (Pipeline automation and data routing).

Fundação Bradesco / Escola Virtual (2025)

- Python Programming Language (53 hours),
- Database Administration & Internet of Things (IoT),
- Data Analysis, Fundamentals,
- Data Modeling in Microsoft Power BI (37 hours).

Udemy (2025)

- Advanced Excel,
- Macro, VBA + Power BI Data Integration.

MIT & Santander Academy

- Internet of Things (IoT),
- Digital Transformation (Connected sensors and cloud telemetry).

IE Business School & Santander

- Data Science & Predictive Modeling Foundations.

Business English

- Santander Open Academy (Technical corporate communication and agile workflows).

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COMPREHENSIVE AGRONOMY & FIELD MANAGEMENT

SENAR EAD (2025)

- Precision Agriculture Professional Series (103h Total),
- Introduction to Precision Agriculture (18h),
- Precision Agriculture in Seeding Systems (17h),
- Precision Agriculture in Crop Spraying & Defensives (17h),
- Precision Agriculture in Grain Harvesting (17h),
- Precision Agriculture in Fertilizer & Corrective Distribution (17h),
- Precision Agriculture in Multi-Crop Environments (17h).

CROP PRODUCTION, PHYTOSANITY & MANAGEMENT

Embrapa (2023)

- Post-Harvest Technology in Fruits and Vegetables,
- Seed and Seedling Production Technology.

UFV (2024 / 2022)

- Good Agricultural Practices (BPA) in Defensives Application,
- II Integrated Pest Management (MIP) Workshop.

UFV & EPAMIG (2024)

- Pest Control in Coffee Crops,
- Conilon Coffee Specialized Production.

Emater-MG (2024)

- Coffea arabica Crop Management (Nutrition, pruning, and phytosanitary metrics).

SUSTAINABILITY & INTEGRATED SYSTEMS

Embrapa (2025 / 2024)

- Crop-Livestock-Forestry Integration Systems (ILPF),
- Degradated Pasture Recovery,
- Bio-inputs Preparation using Organic Waste,
- No-Till Farming Systems (Plantio Direto),
- Soil Management for Integrated Systems,
- Composting,
- Agroforestry Systems (SAF), Residential Aquaponics, Small-Scale Vegetable, Gardening, and Solar Dryer Construction.

BUSINESS & WEB DESIGN BASICS

SENAI Ubá (2014)

- Industrial Apprenticeship in Administrative Processes

Rede Info Ubá / Bradesco

- Web Designer, HTML/CSS/JavaScript Foundations,
- Advanced Office Environments.

Embrapa (2021)

- Foundations of Rural Entrepreneurship.

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